

Take-Home Assignment: The "Contractual Conflict" Challenge

Please read everything carefully before starting the task.

In the construction and engineering industries, a single contradiction in a technical specification can lead to significant financial loss, legal disputes, or structural failure. Your objective is to develop an automated auditor to detect these errors and a benchmarking system to evaluate how effectively an AI performs this high-stakes task.

You are provided with a pdf file: Section 071400 (Bituminous Waterproofing System). There are exactly **ten (10) contradictions** intentionally planted within this document, spanning four categories: **Technical, Responsibility, Liability, and Schedule**.

Materials Provided

- `waterproofing_spec.pdf` — the construction specification under audit (~5 pages)
- `ground_truth.json` — the 10 planted contradictions, with `Statement1`, `Statement2`, and `Reasoning`
- `model_1_output.json` — output from Model 1
- `model_2_output.json` — output from Model 2

Task 1: Contradiction Extraction (Optional)

Build a Python-based discovery tool that identifies internal contradictions within the provided PDF.

Implement an LLM-based algorithm to extract contradictions for **Technical, Responsibility, Liability, and Schedule**.

Keep the output format consistent with the ground truth so that the evaluation is easy for you:

```
{  
  "Statement1":  
  "Statement2":  
  "Reasoning":  
},
```

Note: If time is limited, prioritize Task 2. This task is intended to evaluate Python competency..

Feel free to use any LLM you want. If you need an API key, email rohan.kumaryadav@aecom.com — we will provide one valid for 48 hours.

Task 2: Evaluation & Benchmarking (Mandatory)

The Problem: Model Regression & Validation

In high-stakes document auditing, model updates or architectural changes can lead to regressions, where previously identified errors are missed or new hallucinations are introduced. To ensure reliability, we must implement a rigorous evaluation framework that validates new model results against an established Ground Truth.

The Objective

You are provided with a Ground Truth dataset containing 10 planted contradictions. You are also provided with two sets of LLM outputs (`model_1` and `model_2`) representing different performance tiers. Your goal is to design and implement a methodology to quantify which model is "safer" for production.

Explore and implement a suite of evaluation metrics. At a minimum, your solution should address:

- **Precision & Recall:** How many of the 10 ground truth contradictions were successfully captured, and how many "false alarms" (hallucinations) did the model generate?
- **Matching strategy:** A predicted contradiction will not be worded identically to the ground truth — The same contradiction may appear with different language, wording or section references.. How do you decide when a prediction matches a ground-truth item? Defend the choice.
- **Evaluate on your result from Task 1 (if possible):** If you attempted Task 1, run your own model's output through the same benchmark so we can see how it compares to Models 1 and 2.
- **Report:** Report your findings or evaluation in a way that is more explainable to you. Feel free to be creative here — table, dashboard, narrative, anything that communicates clearly.

What We're Looking For

- Clear evaluation thinking — what to measure, why, and what the trade-offs are.
- Honest interpretation — not just numbers, but what they mean for choosing between the two models in production.
- Open Discussion during interview about how you would approach this problem.

Submission

Please send back a folder or ZIP file via email containing:

- Your evaluation code (e.g. `evaluate.py`)
- The report generated by your code
- A short README on how to run it
- (*If you attempted Task 1*) your extraction code and its output JSON

Time budget: ~2 hours for Task 2 alone, or ~2–3 hours if you also do Task 1.